

Alfred Experiments

Design ICRA

Note: A context switch is defined as any visual attention shift from the primary physical workspace toward a secondary digital information source. Looking from paper to laptop or laptop to paper.

GOAL

Alfred 🏆

1: Combine digital and physical worlds

2: Act as a socratic robotic agent

Therefore ➡

1: Decrease Context Switching

2: Improve Learning

3: Improve Outcome Quality

1: Decrease **Context Switching**:

- Number of switches (2, 3, 4, 5)
- Switch duration (2, 3, 4, 5)
- Resumption lag (time to be productive again) (2, 3, 4, 5)
- Questionnaire preference and context switching (4vs5) (7)

2: Improve **Short term memory => Learning**:

- Δ outcome quality (with vs without help) HW (2, 6)
- Questionnaire learning support and productivity (2, 3, 4) (7)

3: Improve **Outcome Quality**:

- Outcome quality scores HW (2)
- Outcome quality scores input-output GI & DW (3, 4)

EXPERIMENT PROCEDURE

Control (ChatGPT)	Experiment (Alfred)
1. Questionnaire 1 (demographics, pre-assess skills)	1. Questionnaire 1 (demographics, pre-assess skills)
2. HW	2. HW
3. GI	3. GI
4. DW	4. DW
5. DW with Alfred (opposite condition)	5. DW with ChatGPT (opposite condition)
6. HW with no help	6. HW with no help
7. Questionnaire 2 (user preference, perceived context switching, perceived learning support, perceived productivity, feedbacks)	7. Questionnaire 2 (user preference, perceived context switching, perceived learning support, perceived productivity, feedbacks)

MEASUREMENTS

BEHAVIOURS:

- Number of switches
- Switch duration
- Resumption lag (time to be productive again)

RUBRIC	How we will measure them	Type
N switches	Watch footage and count (USE TOOL)	n
Switch duration	Watch footage and measure	min

PERFORMANCE:

- Time
- Outcome quality
- Learning (do task (with vs without help)

RUBRIC	How we will measure them	Type
Time	Start timer at beginning	min
Outcome quality	HW: Task correct? Procedure correct? GI: input & output consistency, creativity and usability DW: recognizability, proportions and completeness	Professor grade form 1 to 10 x3
Short term memory / Learning	Measure outcome quality before (with help) and after (without help) For HW	Grade 1 to 10 before Grade 1 to 10 after For HW

SUBJECTIVE:

- Productivity

- Perceived context switching
- Learning support
- User preference

RUBRIC	How we will measure them	Type
Task Productivity	Questionnaire 2	score
Task Innovation	Questionnaire 2	score
Information Quality	Questionnaire 2	score
Learning Quality	Questionnaire 2	score
Cognitive Demand	Questionnaire 2	score
User Satisfaction	Questionnaire 2	score
User Preference	Questionnaire 2	score
Perceived Context Switching	Questionnaire 2	score
Feedbacks	Questionnaire 2	Open-ended questions

DEMOGRAPHIC:

- Age
- Gender
- Education
- Field of study
- Field of work (if one)
- Experience with robots
- Pre-experiment skill level

RUBRIC	How we will measure them	Type
All	Questionnaire 1	-

QUESTIONNAIRE

Questionnaire 1 (DEMOGRAPHIC):

Demographic

- Age
- Gender
- Education level
- Field of study
- Field of work if any
- Experience with robots
- Experience with AI tools

Pre-experiment skill level

- Self-rated skills in solving math
- Self-rated skills in generate images from physical to digital
- Self-rated skills in drawing

Before answering the following questions Please take a moment to think back over your overall experience using this system across the three tasks: (1) Homework, (2) Generate Image, and (3) Drawing. Consider what you did, how the system supported you, how you felt during the tasks, and what you experienced while working with it. Also, briefly recall the final Homework task completed without any assistance, and consider how your experience with the system may have influenced your understanding, confidence, or ability to work independently. Please answer the following questions based on your overall experience with the system across the study, rather than on a single task.

Questionnaire 2 (SUBJECTIVE Control): RANDOMISE QUESTIONS- MAKE A GRID

Task Productivity (2, 3, 4, (6))

- ChatGPT saves me time
- ChatGPT increases my productivity
- ChatGPT allows me to accomplish more work than would otherwise be possible
- ChatGPT improves the quality of my work. -> *Learning Quality*
- ChatGPT enables me to spend more time on productive activities

Task Innovation (2, 3, 4, (6))

- ChatGPT helps me come up with new ideas
- ChatGPT helps me try out innovative ideas
- ChatGPT helps me find new ways to improve my task performance

Information Quality (2, 3, 4, (6)) SPLIT QUESTION

- ChatGPT clearly states the goal for the tasks
- ChatGPT clearly states the procedure for the tasks
- ChatGPT guidance is an appropriate support to my learning process

Learning Quality (2, 3, 4, (6)) SIMPLIFY WORDING

- ChatGPT has the potential to make the learning process more engaging and motivating
- ChatGPT supports critical aspects of my learning
- ChatGPT facilitate my learning by engaging me with relevant content
- Using ChatGPT, I am able to learn through several modalities (such as audio, visual, video, and text -based formats)
- The environment is enjoyable for learning overall
- I find that ChatGPT is a good educational portal that improves my learning
- ChatGPT uses technologies in a creative way that go beyond traditional ways of learning
- ChatGPT encourages me to continue learning

Cognitive Demand (2, 3, 4, (6))

- Overall the tasks were mentally demanding
- Overall the tasks were physically demanding

User Satisfaction (2, 3, 4, (6))

- ChatGPT improves my task satisfaction
- ChatGPT helps me meet my needs
- ChatGPT helps me create value for my task understanding
- ChatGPT helps me create value for my task outcome

User Preferences (4 vs 5)

- Overall, I preferred using ChatGPT over Alfred for the drawing task.

Before answering the following questions Please take a moment to think back over your overall experience using this system across the three tasks: (1) Homework, (2) Generate Image, and (3) Drawing. Recall how you worked with the system, how you felt, how it supported your work, and how you experienced the interaction with it. Also remember the second Drawing task, which you completed using the other system, and use this experience as a point of comparison. When answering the following questions, consider your experience across all three tasks with this system, while also keeping in mind how your experience differed when using the other system during the second Drawing task. Please answer based on your actual experiences during the study, rather than on what you expected either system to be like.

Perceived Context Switching (2, 3, 4, (5, 6))

- I was able to maintain my attention on my physical work while using the system.
- The system allowed me to access information without interrupting my concentration.
- I could use the system without frequently shifting my attention away from my work.
- The system helped me maintain a continuous workflow between receiving assistance and performing the task.

Feedbacks

- What aspects of ChatGPT were most helpful during the tasks? Please explain why.
- What aspects of ChatGPT made the tasks more difficult or less effective?
- After experiencing both systems, what do you think were the biggest differences between ChatGPT and Alfred?
- Which system would you prefer to use for similar desk-based learning or creative tasks, and why?
- Do you have any suggestions that could improve Alfred or make it more useful during these tasks?
- Is there anything else you would like to share about your experience during the study?

Questionnaire 2 (SUBJECTIVE Experiment): SAME AS BEFORE BUT FOR ALFRED

Task Productivity

- Alfred saves me time.
- Alfred increases my productivity.
- Alfred allows me to accomplish more work than would otherwise be possible.
- Alfred improves the quality of my work.
- Alfred enables me to spend more time on productive activities.

Task Innovation

- Alfred helps me come up with new ideas.
- Alfred helps me try out innovative ideas.
- Alfred helps me find new ways to improve my task performance.

Information Quality

- Alfred clearly states the goal and procedure for the tasks.
- Alfred guidance is an appropriate support to my learning process.

Learning Quality

- Alfred has the potential to make the learning process more engaging and motivating.
- Alfred supports critical aspects of my learning.
- Alfred facilitates my learning by engaging me with relevant content.
- Using Alfred, I am able to learn through several modalities (audio, visual, video, and text-based formats).
- The environment is conducive and enjoyable for learning overall.
- I find that Alfred is a good educational portal that improves my learning.
- Alfred uses technologies in a creative way that go beyond traditional ways of learning.
- Alfred encourages me to continue learning.

Cognitive Demand

- How mentally demanding was the task?

- How physically demanding was the task?
- How hurried or rushed was the pace of the task?

Perceived Context Switching

- I was able to maintain my attention on my physical work while using Alfred.
- Alfred allowed me to access information without interrupting my concentration.
- I could use Alfred without frequently shifting my attention away from my work.
- Alfred helped me maintain a continuous workflow between receiving assistance and performing the task.

User Satisfaction

- Alfred improves my task satisfaction.
- Alfred helps me meet my needs.
- Alfred helps me create value for my task understanding.
- Alfred helps me create value for my task outcome.

User Preferences

- Overall, I preferred using Alfred over ChatGPT for the drawing task.
- If I had to complete similar drawing tasks again, I would choose Alfred.
- I would recommend Alfred over ChatGPT for similar drawing tasks.
- Overall, I would prefer to use Alfred rather than ChatGPT for similar desk-based tasks in the future.

Feedback

- What aspects of Alfred were most helpful during the tasks? Please explain why.
- What aspects of Alfred made the tasks more difficult or less effective?
- After experiencing both systems, what do you think were the biggest differences between Alfred and ChatGPT?
- Which system would you prefer to use for similar desk-based learning or creative tasks, and why?
- Do you have any suggestions that could improve Alfred or make it more useful during these tasks?
- Is there anything else you would like to share about your experience during the study?

ID

Participant

Condition : Experiment or Control

Email

Date

(C:\Users\grego\Pictures\Camera Roll\Participants)

Video : 1 = we have in user folder 0 = need to put in user folder

HW : 1 = we have in user folder 0 = need to put in user folder

HW no help : 1 = we have in user folder 0 = need to put in user folder

DW in : 1 = we have in user folder 0 = need to put in user folder

DW out : 1 = we have in user folder 0 = need to put in user folder

DW out opposite : 1 = we have in user folder 0 = need to put in user folder

GI in : 1 = we have in user folder 0 = need to put in user folder

GI out : 1 = we have in user folder 0 = need to put in user folder

t HM : time from task delivered to user finish

t GI : time from task delivered to user finish

t DW : time from task delivered to user finish

Actual t DW : time from task started drawing to user finish

t DW opposite : time from task delivered to user finish

Actual t DW opposite : time from task started drawing to user finish

t HW no help : time from task delivered to user finish

Number Switches HW (HW nS) : tot number of switches (python code)

Avg. Paper Look HW (HW avg paper) : avg. time the user looks at paper before looking at monitor

Avg. Monitor Look HW (HW avg monitor) : avg. time the user looks at monitor before looking at paper

t on paper HW (HW t paper) : tot time looking at paper

Number Switches GI : tot number of switches (python code)

Avg. Paper Look GI : avg. time the user looks at paper before looking at monitor

Avg. Monitor Look GI : avg. time the user looks at monitor before looking at paper

t on paper GI : tot time looking at paper

Number Switches DW : tot number of switches (python code)

Avg. Paper Look DW : avg. time the user looks at paper before looking at monitor

Avg. Monitor Look DW : avg. time the user looks at monitor before looking at paper

t on paper DW : tot time looking at paper

Grade Math : grade given by AI professor to result

Grade Math With No Help : grade given by AI professor to result

Grade Generate Image input-output : grade given by AI professor to result

n Generation: How many interaction of the image where done

Grade Input Draw : grade given by AI professor to result

Grade Output Draw : grade given by AI professor to result

Grade Output Draw Opposite Condition : grade given by AI professor to result

Age, Gender, ...

LITERATURE REVIEW

For Questionnaires:

- The development of a tool for measuring the perceived impact of information technology on work
- What drives a successful e-Learning? An empirical investigation of the critical factors influencing learner satisfaction
- Why is it so hard to do my work? The challenge of attention residue when switching between work tasks
- Perceptions of learning management system quality, satisfaction, and usage: Differences among students of the arts
- Technology and instructor dimensions, e-learning satisfaction, and academic performance of distance students in Ghana

For SoA:

- [AI advice suppresses people's willingness to say "I don't know", even when the advice is wrong and accuracy is incentivized](#)
- [Is AI making us stupid?](#)
- [AI's impact on cognitive ability: MIT study reveals more troubling data](#)
- [The Pen Is Mightier Than the Keyboard: Advantages of Longhand Over Laptop Note Taking](#)
- [The ICAP Framework: Linking Cognitive Engagement to Active Learning Outcomes](#)
- [Cognitive Offload](#)
- [Structured Prompts, Better Outcomes? Exploring the Effects of a Structured Interface with ChatGPT in a Graduate Robotics Course](#)
- [Inside AI Podcast \(4Cs\)](#)
- [4C blog MIT](#)
- [The 4Cs Learning Model in Teacher Professional Development Program](#)
- [Socratic Agent](#)
- [Enhancing Critical Thinking in Education by means of a Socratic Chatbot](#)

- [Francesco Mondada](#)
- [IllumiShare](#)
- [AppleElegnt](#)
- [Simply Draw-Sing-Guitar-Piano](#)